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United States Patent	12409759
Kind Code	B2
Date of Patent	September 09, 2025
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Systems and methods to monitor seats for diagnostics, prognostics and degradation

Abstract

A method is provided for detecting a fault in a gear system for repositioning a seat assembly in an automotive vehicle. The gear system includes a gear train operatively coupled to a motor. The method includes the steps of measuring a current drawn by the motor to reposition the seat assembly, recording the current over time as a current waveform, converting the current waveform into a frequency domain, determining a base frequency of the current waveform based on the converted current waveform, calculating a power spectrum density at the base frequency, calculating a total band power of the power spectrum density, and determining whether the total band power is greater than a predetermined total band power threshold.

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Appl. No.:	18/284965
Filed (or PCT Filed):	March 30, 2022
PCT No.:	PCT/US2022/022422
PCT Pub. No.:	WO2022/212427
PCT Pub. Date:	October 06, 2022

Prior Publication Data

Document Identifier	Publication Date
US 20240181937 A1	Jun. 06, 2024

Related U.S. Application Data

us-provisional-application US 63167938 20210330

Publication Classification

Int. Cl.: B60N2/02 (20060101); G01M13/021 (20190101); G01R23/18 (20060101)

U.S. Cl.:

CPC B60N2/0244 (20130101); B60N2/02253 (20230801); G01M13/021 (20130101); G01R23/18 (20130101);

Field of Classification Search

CPC: B60N (2/0244); B60N (2/02253); G01M (13/021); G01R (23/18)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
6507789	12/2002	Reddy et al.	N/A	N/A
7019660	12/2005	Seto et al.	N/A	N/A
10833622	12/2019	Li et al.	N/A	N/A
10910977	12/2020	Barlini	N/A	N/A
2003/0080699	12/2002	Rumney	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
106950058	12/2016	CN	N/A
1489401	12/2003	EP	N/A
2743670	12/2013	EP	N/A
20200081591	12/2019	KR	N/A
2003002256	12/2002	WO	N/A

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application claims priority to U.S. Provisional Application 63/167,938, filed on Mar. 30, 2021, the disclosure of which is hereby incorporated by reference in its entirety.

BACKGROUND OF THE INVENTION

1. Field of the Invention

(1) The present invention relates to a seat assembly having a gear train. More particularly, the invention relates to methods of detecting faults within the gear train of the seat assembly.

2. Description of Related Art

(2) Seat assemblies in automotive vehicles typically include systems to control the movement and positioning of the seat assemblies. Over time, performance of the systems degrades due to repetitive operation or aging effects, which may impede the smooth transition between the seat configurations, alter the positioning of the seat assembly, and/or diminish proper functioning of the seat assembly. In addition, expected and unexpected impacts can change various system components. There is a need for a system that can detect variations that occur over time so that the system may be repaired before the degradation becomes noticeable to the user and/or the system becomes non-operational.

SUMMARY OF THE INVENTION

(3) According to one embodiment, there is provided a method for detecting a fault in a gear system for repositioning a seat assembly in an automotive vehicle. The gear system includes a gear train operatively coupled to a motor. The method comprises the steps of measuring a current drawn by the motor to reposition the seat assembly, recording the current over time as a current waveform, converting the current waveform into a frequency domain, determining a base frequency of the current waveform based on the converted current waveform, calculating a power spectrum density at the base frequency, calculating a total band power of the power spectrum density, and determining whether the total band power is greater than a predetermined total band power threshold.

(4) According to another embodiment, there is provided a method for detecting a fault in a gear system for repositioning a seat assembly in an automotive vehicle. The gear system includes a gear train operatively coupled to a motor. The method comprises the steps of measuring a current drawn by the motor to reposition the seat assembly, recording the current over time as a current waveform, using a continuous wavelet transformation process to calculate a signature of the current waveform, and determining whether the signature is greater than a predetermined signature threshold.

(5) According to another embodiment, there is provided a method for detecting a fault in a gear system for repositioning a seat assembly in an automotive vehicle. The gear system includes a gear train operatively coupled to a motor. The method comprises the steps of applying a voltage pulse to the motor, measuring a current drawn by the motor while the voltage pulse is applied to the motor, recording the current over time as a current waveform, determining a rise time in the current waveform for the current to reach a predetermined current amplitude, and determining whether the rise time is greater than a predetermined rise time threshold.

(6) According to another embodiment, there is provided a method for detecting a fault in a gear system for repositioning a seat assembly in an automotive vehicle. The gear system includes a gear train operatively coupled to a motor having a drive shaft. The method comprises the steps of measuring a hall effect signal from the drive shaft indicating a rotational position and a rotational speed of the drive shaft over time, comparing the hall effect signal to an expected hall effect signal for the motor, and determining whether the hall effect signal is different from the expected hall effect signal.

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) Advantages of the present invention will be readily appreciated as the same becomes better understood by reference to the following detailed description when considered in connection with the accompanying drawings wherein:
- (2) FIG. 1 is a perspective view of a seat assembly having a gear system, according to one embodiment of the present invention:
- (3) FIG. 2 is a perspective view of the gear system of FIG. 1:
- (4) FIG. 3 is an enlarged schematic representation of meshed gear teeth within the gear train of FIG. 2:
- (5) FIG. 4 is an enlarged schematic representation of the meshed gear teeth in FIG. 3, after the gear teeth become damaged:
- (6) FIG. 5 is a graphical representation of current drawn by a motor during operation of the gear train of FIG. 2:
- (7) FIG. 6 is a graphical representation of demanded torque by the motor during operation of the gear train of FIG. 2:
- (8) FIG. 7 is a graphical representation of drive shaft speed in comparison to applied drive force for the motor of FIG. 2:
- (9) FIG. 8 is a graphical representation of hall effect sensor feedback in response to provided power to the motor when the gear train of FIG. 2 is healthy:
- (10) FIG. 9 is a graphical representation of hall effect sensor feedback in response to the motor stalling when the gear train of FIG. 2 is healthy:
- (11) FIG. 10 is a graphical representation of hall effect sensor feedback in response to provided power to the motor when the gear train of FIG. 2 is unhealthy:
- (12) FIG. 11 is a graphical representation of hall effect sensor feedback in response to the motor stalling when the gear train of FIG. 2 is unhealthy:
- (13) FIG. 12 is a graphical representation of the current measurements of FIG. 5 converted to the frequency domain:
- (14) FIG. 13 is a graphical representation of calculated power spectrum densities at the base frequencies of FIG. 12 for a healthy gear train:
- (15) FIG. 14 is a graphical representation of calculated power spectrum densities at the base frequencies of FIG. 12 for an unhealthy gear train:
- (16) FIG. 15 is a flow chart of a first method for detecting an unhealthy gear train, according to one embodiment of the present invention:
- (17) FIG. 16 is a flow chart of a second method for detecting an unhealthy gear train, according to another embodiment of the present invention:
- (18) FIG. 17 is a flow chart of a third method for detecting an unhealthy gear train, according to another embodiment of the present invention:
- (19) FIG. 18 is a graphical representation of a voltage pulse, according to another embodiment of the present invention:
- (20) FIG. 19 is a graphical representation of current drawn by the motor of FIG. 2 in response to the voltage pulse of FIG. 18:
- (21) FIGS. 20A-B show a flow chart of a fourth method for detecting an unhealthy gear train, according to another embodiment of the present invention: and
- (22) FIGS. 21A-B show a flow chart of a fifth method for detecting an unhealthy gear train, according to another embodiment of the present invention.

DETAILED DESCRIPTION OF THE EMBODIMENTS

(23) The present invention relates to systems and methods for detecting faults in the operation of seat assemblies **10** in automotive vehicles. Directional references employed or shown in the description, figures, or claims, such as top, bottom, upper, lower, upward, downward, lengthwise,

widthwise, left, right, and the like, are relative terms employed for ease of description and are not intended to limit the scope of the invention in any respect. Referring to the Figures, like numerals indicate like or corresponding parts throughout the several views.

(24) As depicted in FIG. 1, the seat assembly **10** includes a seat back **12**, a seat cushion **14**, a seat base **16**, a pair of adjuster assemblies **18**, and a linkage assembly **20**. The seat back **12** is rotatably coupled to the seat cushion **14**. The linkage assembly **20** pivotably couples the seat cushion **14** to the seat base **16**. The seat base **16** is transposable fore and aft along the pair of adjuster assemblies **18**.

(25) The seat assembly **10** also includes an electronic control unit (ECU) **22** and one or more gear systems **24**. Each gear system **24** includes a gear train **26** operatively coupled with one or more motors **29**. The ECU **22** controls the movement and positioning of the seat assembly **10** by controlling the motors **29** which drive the gear trains **26**. The ECU **22** not only controls the movement and positioning of the seat assembly **10**, but it also monitors these activities to ensure that they are working properly over time.

(26) The exemplary gear train **26** shown in FIG. 2 includes a drive gear **30**, a plurality of driven gears **31**, **32** and a rack **33**. Each of the gears **30**, **31**, **32** and the rack **33** includes teeth **30a**, **31a**, **32a**, **33a** that meshingly engage with each other. The driven gears **31**, **32** are each coupled to a respective shaft **34**, **35** that is operatively coupled to other components of the seat assembly **10**, e.g., the linkage assembly **20**. The motor **29** includes a drive shaft **36** fixedly coupled to the drive gear **30**. The gear shafts **34**, **35** and the drive shaft **36** are rotationally supported by one or more bearings, thrust washers, and the like.

(27) The gear train **26** is configured to generally lock in place when in non-operating conditions. Maintaining proper locking of the gears **30-33** avoids unexpected movement of the seat assembly **10**.

(28) The seat assembly **10** includes a hall effect sensor **38** that is operatively connected to the drive shaft **36**. The hall effect sensor **38** outputs a hall effect signal **42** comprising a series of hall effect pulses **44** (see FIG. 8) indicative of the rotational position and the rotational speed of the drive shaft **36**. Because the rotational position of the drive shaft **36** is related to movement of the gear train **26** and the resulting movement of the seat assembly **10**, the ECU **22** can be configured to predict the resulting movement of the seat assembly **10** based in part on the hall effect signal **42**.

(29) Over time, the components of the gear system **24** can degrade or become faulty. For example, the gear teeth **30a**, **31a**, **32a**, **33a**, bearings or shafts **34**, **35**, **36** may become worn or damaged or become misaligned. These defects can alter or impede the performance of the gear train **26**. For example, the gear system **24** may become inoperable or the gear train **26** may not lock properly. The defects also may increase the amount of noise or vibration in the seat assembly **10** or cause unexpected movements.

(30) A gear system **24** that is operating as expected and is free of defects is generically described as a healthy gear system **24**. In contrast, a gear system **24** having defects is generically described as an unhealthy gear system **24**.

(31) FIG. 3 shows the engagement of the gear teeth **30a**, **31a** when the gears **30**, **31** are free of defects, while FIG. 4 shows the engagement of the gear teeth **30a**, **31a** when the gears **30**, **31** are worn or damaged. As shown in FIG. 3, the gear teeth **30a**, **31a** are typically configured such that there is a predefined amount of backlash **46** between the teeth **30a**, **31a**. As shown in FIG. 4, wear or damage to the gear teeth **30a**, **31a** may alter the spacing and increase the backlash **46** between the teeth **30a**, **31a**, which translates into loss motion between the gears **30**, **31**. Damaged gears **30**, **31** may slip under load or shift relative to each other so that the gears **30**, **31** may rotate relative to each other when in non-operating conditions. Movement of the gear train **26** during non-operating conditions is undesirable and is described as improper locking.

(32) Each gear **30**, **31** has an operating pitch circle **48**, **50**. In a healthy gear train **26**, the operating pitch circles **48**, **50** are typically tangent to each other, as shown in FIG. 3. Bearing wear and shaft

34, 35, 36 wear may cause the gears **30, 31** to separate or shift relative to each other, forming a gap **52** between the operating pitch circles **48, 50** of the gears **30, 31**, as shown in FIG. 4. In addition, damaged bearings can cause a “wobbly axis” where the drive shaft **36** or the gear shafts **34, 35** wobble within the opening of the bearing around the longitudinal axis of the bearing.

(33) The mechanical connections between the gears **30, 31, 32** and the respective mounting shafts **36, 34, 35** may loosen over time to cause a “loose axis” where a gear **30-32** will rotate relative to a mounting shaft **36, 34, 35**. A gear **30-32** that is loose on a mounting shaft **36, 34, 35** can result in improper locking in the gear train **26** and can result in damage to the gear teeth **30a-33a**. Further, the improper locking can cause unexpected movement of the seat assembly **10** during adverse conditions.

(34) The ECU **22** is configured to monitor the current drawn by the motor **29** during operation of the gear train **26**. The ECU **22** also records the current over time, as reflected in the exemplary current waveform **54** shown in FIG. 5. When the motor **29** initially receives power, the motor **29** begins to rotate the drive shaft **36** causing the drive gear **30** to rotate. The current drawn by the motor **29** rapidly increases as the motor **29** initiates rotation of the drive shaft **36** and all of the gears **30-32** and equipment tied to the gear train **26**, as reflected in portion A of the current waveform **54** in FIG. 5. After the gear train **26** starts moving, less power is needed to drive the gear train **26**. Therefore, the current will begin to decrease after it reaches a peak, **56**, as reflected in portion B of the current waveform **54**. Portion C shows the current waveform **54** during steady state conditions. As shown, the steady state current forms ripples that reflect torsional vibrations in the gear system **24**. The gears **30-32** and motor **29** cause torsional vibrations in the gear system **24** that are related to a number of factors including the speed with which each gear **30-32** is rotating, the number of gear teeth **30a-32a** in each gear **30-32**, and the speed of rotation of the mounting shafts **36, 34, 35**. The torsional vibrations include vibrational components having base frequencies **F29-F32**, which reflect the operating frequencies of the motor **29** and the gears **30-32** during steady state conditions. The base frequency **F29** of the motor **29** is typically larger than the base frequencies **F30-F32** of the individual gears **30-32** of the gear train **26**. Thus, the ripples in the current waveform **54** include frequency components **F29-F32** of the motor **29** and gears **30-32** of the gear train **26**. Portion D of FIG. 5 shows the current waveform **54** when the gear train **26** encounters an expected or unexpected impact.

(35) FIG. 6 shows a graphical representation of an exemplary demanded torque **58** applied by the motor **29** during steady state conditions. The torque **58** forms torque ripples, which reflect the torsional vibrations in the gear system **24**. Similar to the ripples in the current waveform **54**, these torque ripples include frequency components **F29-F32** of the motor **29** and the gears **30-32** of the gear train **26**.

(36) Defects in the gear system **24** are reflected in the current waveform **54** drawn by the motor **29** and the demanded torque **58** applied by the motor **29** during steady state conditions. For example, FIG. 5 illustrates some differences in the current waveform **54** when the gear system **24** is healthy versus the current waveform **54'** when the gear system **24** is unhealthy. As shown, the maximum amplitude peak **56** of the current waveform **54** of the healthy gear system **24** is larger than the maximum amplitude peak **56'** of the current waveform **54'** of the unhealthy gear system **24**. In addition, the rise time **dt1** to reach a maximum amplitude peak **56** for the healthy gear system **24** is shorter than the rise time **dt2** to reach a maximum amplitude peak **56'** for the unhealthy gear system **24**.

(37) Further, during steady state conditions, as reflected in portion C of FIG. 5, the current waveform **54'** driving the unhealthy gear system **24** is not as consistent as the current waveform **54** driving the healthy gear system **24**, and has a greater amplitude variation **60** than the amplitude variation **62** of the healthy current waveform **54**. It is to be appreciated that the actual current waveforms **54, 54'** drawn by the motor **29** during operation of the gear system **24** will vary based in part on the specific gear train **26**, the specific motor **29**, and factors such as specific defects present

in the unhealthy gear system **24**.

(38) Certain motors **29** are powered using a pulse width modulated (PWM) signal that can vary from 0% duty cycle (i.e., no signal is provided) to 100% duty cycle (i.e., full power is provided). The drive force provided to the motor **29** is related to the duty cycle of the PWM signal. FIG. 7 shows an exemplary transfer function **64** between the rotational speed of the drive shaft **36** in relation to the drive force provided to the motor **29** with a healthy gear system **24**. As depicted, the rotational speed of the drive shaft **36** increases as the PWM duty cycle increases, while the slope of the curve decreases with increasing PWM duty cycles. When the gear train **26** has one or more wobbly axes of shafts **36**, **34**, **35**, the resulting transfer function **66** is similar to the normal transfer function **64**, but includes oscillations. In contrast, when the gear train has one or more loose axes of shafts **36**, **34**, **35**, the transfer function **68** reflects a lower rotational speed of the drive shaft **36** with increasing drive force, and initially includes oscillations.

(39) Referring to FIG. 8, when the gear system **24** is healthy, exemplary hall effect pulses **44** have a consistent pulse period P (i.e., the amount of time between successive transitions from low voltage LV to high voltage HV). In contrast, the pulse period P in an unhealthy gear system **24** may fluctuate over time, as illustrated in FIG. 10. Thus, variation in the pulse periods P under constant PWM power may be indicative of faults in gear system **24**, such as one or more loose or wobbly axes of shafts **36**, **34**, **35**, or unacceptable backlash **46**.

(40) FIG. 9 illustrates exemplary hall effect pulses **44** when the drive shaft **36** stops rotating, such as when PWM power is terminated to the motor **29** or the seat assembly **10** impacts an expected or unexpected obstruction. When the power is terminated or the seat assembly **10** impacts an obstruction at time **70**, the hall effect pulses **44** will terminate shortly thereafter.

(41) In contrast and as illustrated in FIG. 11, with an unhealthy gear system **24**, the backlash **46** in the gear train **26** and/or defects such as one or more wobbly axes or loose axes of shafts **36**, **34**, **35** may cause additional oscillatory rotational movement of the drive shaft **36** in both forward and rearward rotational directions after the power is terminated or the seat assembly **10** impacts an obstruction, resulting in more hall effect pulses **44** appearing after time **70** compared to the number of hall effect pulses **44** appearing after time **70** with a healthy gear system **24**. A stop threshold **76** can be experimentally determined to identify unhealthy gear systems **24** when the amount of hall effect pulses **44** detected after impact **70** is greater than the stop threshold **76**.

(42) Defects in the gear system **24** can be detected by analyzing the current waveform **54** using motor current signature analysis (MCSA) to isolate and evaluate changes in the frequency components F29-F32. A method for determining whether there are any faults in the gear system **24** according to a first embodiment of the present invention is shown in FIGS. 5, 6, and 12 through 15. Referring to FIG. 15, the ECU **22** initially provides power to the motor **29** (step **88**) and records the motor current waveform **54**, **54'** (step **90**). The ECU **22** then filters noise from the steady state portion C of the current waveform **54**, **54'** (step **92**), and converts the filtered current waveform **54**, **54'** into the frequency domain **94** (step **96**). FIG. 12 shows a graphical representation of the current waveform **54**, **54'** converted in the frequency domain **94**. The base frequencies F29-F32 shown in FIG. 12 correspond to the base frequencies F29-F32 in the torsional vibrations caused by the motor **29** and the gears **30-32**, respectively. Because the frequency components F29-F32 are also included in the torque ripples, the ECU **22** alternatively may convert the demanded torque **58** into the frequency domain **94** rather than converting the current waveform **54**, **54'** into the frequency domain **94** at step **96**.

(43) The ECU **22** then calculates the power spectrum density **98** at the base frequencies F29-F32 and at the harmonics of the base frequencies F29-F32 (FIG. 15, step **100**). Referring to FIG. 13, the power spectrum density **98** is concentrated near the base frequencies F29-F32 when the gear system **24** is healthy. In contrast, when the gear system **24** is unhealthy, the power spectrum density **98'** will accumulate around the base frequencies F29-F32 and around harmonics of the base frequencies F29-F32, as shown in FIG. 14.

(44) Referring back to FIG. 15, the ECU 22 then calculates the total band power at the base frequencies F29-F32 and the harmonics of the base frequencies F29-F32 (step 102), and compares the calculated total band power to a predetermined total band power threshold (step 104). If the calculated total band power is less than or equal to the total band power threshold, then the gear system 24 is considered healthy and the ECU 22 continues operating the gear system 24 (step 106). If the calculated total band power is greater than the total band power threshold, then the gear system 24 is considered unhealthy and the ECU 22 stores a corresponding fault code in memory (step 108), and notifies the occupant that a fault has been detected (step 110). The ECU 22 then continues operation of the gear system 24 (step 106).

(45) A method for determining whether there are any faults in the gear system 24 according to a second embodiment of the present invention is shown in FIG. 16. In the second method, the ECU 22 initially provides power to the motor 29 (step 114) and records the motor current waveform 54, 54' (step 116). The ECU 22 then filters noise from the current waveform 54, 54' (step 118) and transforms the filtered current waveform 54, 54' using continuous wavelet transformation to produce a calculated signature (step 120). Continuous wavelet transformation uses a small specifically designed wave (which is also stretchable) to convolute with the current waveform 54, 54' being analyzed to extract hidden abnormal signatures embedded in transient and local data.

(46) The ECU 22 compares the calculated signature with a predetermined signature threshold (step 122). The signature threshold is determined by comparing the signatures of the healthy and unhealthy gear systems 24. If the calculated signature is less than the signature threshold, then the gear system 24 is considered healthy and the ECU 22 continues operating the gear system 24 (step 124). If the calculated signature is greater than the signature threshold, then the gear system 24 is considered unhealthy and the ECU 22 stores a corresponding fault code in memory (step 126), and notifies the occupant that a fault has been detected (step 128). The ECU 22 then continues operation of the gear system 24 (step 124).

(47) A method for determining whether there are any faults in the gear system 24 according to a third embodiment of the present invention is shown in FIGS. 17 through 19. Referring to FIG. 17, the ECU 22 initially provides a voltage pulse VP to the motor 29 (step 132) while the gear train 26 is in a locked condition. The voltage pulse VP has a voltage amplitude V1 and a time period T1 that is insufficient to cause unexpected movement of the seat assembly 10. In one embodiment, a controlled voltage pulse VP having a voltage amplitude V1 of about 7 volts is provided by the ECU 22 across the terminals of the motor 29 for a time period T1 of approximately 0.5 seconds, as shown in FIG. 18.

(48) Referring back to FIG. 17, the ECU 22 records the resulting motor current waveform 134, 134' (step 136) and filters noise from the current waveform 134, 134' (step 138). FIG. 19 shows an exemplary current waveform 134 when the gear train 26 is properly locked, and an exemplary current waveform 134' when the gear train 26 is improperly locked. Referring to FIGS. 17 and 19, the ECU 22 measures the rise time dt3, dt4 between initialization of the controlled voltage pulse VP and the time that the current waveform 134, 134' reaches a predetermined amplitude 140 (step 142). The ECU 22 compares the measured rise time dt3, dt4 to a rise time threshold RT (step 144). The predetermined current amplitude 140 and the rise time threshold RT are selected based on experimental data comparing a healthy gear system 24 with an unhealthy gear system 24. Alternatively, the current amplitude 140 and the rise time threshold RT can be determined based in part on a prior evaluation of the gear system 24 by the ECU 22.

(49) Referring to FIG. 17, if the measured rise time dt3, dt4 is less than or equal to the rise time threshold RT, then the gear system 24 is considered healthy and the ECU 22 continues operating the gear system 24 (step 146). If the measured rise time dt3, dt4 is greater than the rise time threshold RT, then the locking of the gear train 26 is not proper and the ECU 22 stores a corresponding fault code in memory (step 148), and notifies the occupant that a fault has been detected (step 150). The ECU 22 then continues operation of the gear system 24 (step 146).

(50) A method for determining whether there are any faults in the gear system **24** according to a fourth embodiment of the present invention is shown in FIGS. **20A-B**. In the fourth method, the ECU **22** initially provides power to the motor **29** (step **152**) and records the hall effect pulses **44** received from the hall effect sensor **38** in response to rotation of the drive shaft **36** (step **154**). Next, the ECU **22** evaluates the hall effect pulses **44** (step **156**). If the hall effect pulses **44** are irregular, then the gear system **24** is considered unhealthy, and the ECU **22** stores a corresponding fault code in memory (step **158**) and notifies the occupant that a fault has been detected (step **160**). After notifying the occupant about the fault, the ECU **22** continues to operate the gear system **24** (step **162**). The ECU **22** also continues to operate the gear system **24** if at step **156** it did not detect irregular hall effect pulses **44**.

(51) While the ECU **22** operates the gear system **24** (step **162**), it monitors for impacts (step **164**) and waits until power to the motor **29** is terminated (step **166**). If the ECU **22** detects an impact, then the ECU **22** counts the hall effect pulses **44** received after the impact was detected (step **168**) and determines whether the counted hall effect pulses **44** exceeds a predetermined impact threshold (step **170**). If the counted hall effect pulses **44** is greater than the impact threshold, then the gear system **24** is considered unhealthy and the ECU **22** stores a corresponding fault code in memory (step **172**), and notifies the occupant that a fault has been detected (step **174**). If the counted hall effect pulses **44** is less than or equal to the impact threshold, then no further action is required by the ECU **22** (step **176**).

(52) If the ECU **22** terminates power to the motor **29** (step **166**), then the ECU **22** counts the hall effect pulses **44** received after power was terminated to the motor **29** (step **178**), and determines whether the counted hall effect pulses **44** exceeds the predetermined stop threshold (step **180**). If the counted hall effect pulses **44** is greater than the stop threshold, then the gear system **24** is considered unhealthy and the ECU **22** stores a corresponding fault code in memory (step **182**), and notifies the occupant that a fault has been detected (step **184**). If the counted hall effect pulses **44** is less than or equal to the stop threshold, then no further action is required by the ECU **22** (step **176**).

(53) Each of the methods for evaluating the condition of the gear system **24** disclosed above may be repeated at any desired interval without varying the scope of the invention. For example, an exemplary method for notifying occupants of the seat assemblies **10** when to bring the automotive vehicle into service is shown in FIGS. **21A-B**. Initially, all counters, including a counter for tracking the number of ignition cycles run between system maintenance and the counters used to track various defects in the seat assembly **10**, are reset to zero (step **188**). The ECU **22** waits for the start of a new ignition cycle (step **190**) before it increments the ignition cycle counter (step **192**). The ECU **22** then determines whether the ignition cycle counter exceeds a threshold (step **194**). If the ignition cycle counter exceeds the threshold, then the ECU **22** alerts the occupant that it is time to bring the automotive vehicle in for system maintenance (step **196**).

(54) The ECU **22** then checks the gear system **24** for any faults (step **198**). For example, the ECU **22** may run one or more of the methods disclosed above. The ECU **22** determines if any faults were detected (step **200**). If any faults were detected, the ECU **22** identifies the fault type (step **202**) and increments the counter for that fault type (step **204**). The ECU **22** then checks if the counter for the fault type exceeds a threshold (step **206**). If the counter for the fault type exceeds the threshold, then the ECU **22** notifies the occupant that the fault was detected (step **208**) so that the occupant may bring the automotive vehicle in for service. The ECU **22** then waits for the next ignition cycle before running the process again.

(55) The invention has been described in an illustrative manner, and it is to be understood that the terminology, which has been used, is intended to be in the nature of words of description rather than of limitation. Many modifications and variations of the present invention are possible in light of the above teachings. It is, therefore, to be understood that within the scope of the appended claims, the invention may be practiced other than as specifically described.

Claims

1. A method for detecting a fault in a gear system for repositioning a seat assembly in an automotive vehicle, wherein the gear system includes a gear train operatively coupled to a motor, the method comprising the steps of: measuring a current drawn by the motor to reposition the seat assembly; recording the current over time as a current waveform; converting the current waveform into a frequency domain; determining a base frequency of the current waveform based on the converted current waveform; calculating a power spectrum density at the base frequency; calculating a total band power of the power spectrum density; and determining whether the total band power is greater than a predetermined total band power threshold.
2. The method as set forth in claim 1, wherein if it is determined that the total band power is greater than the predetermined total band power threshold, recording an indication of a fault in memory.
3. The method as set forth in claim 1, wherein if it is determined that the total band power is greater than the predetermined total band power threshold, providing a notification that a fault has been detected.
4. The method as set forth in claim 1, further comprising the steps of: monitoring a number of times that the total band power is greater than the predetermined total band power threshold; and if the number exceeds a count threshold, providing a notification that a fault has been detected.
5. The method as set forth in claim 1, further comprising the step of filtering noise from the current waveform prior to transforming the current waveform into the frequency domain.
6. A method for detecting a fault in a gear system for repositioning a seat assembly in an automotive vehicle, wherein the gear system includes a gear train operatively coupled to a motor, the method comprising the steps of: measuring a current drawn by the motor to reposition the seat assembly; recording the current over time as a current waveform; using a continuous wavelet transformation process to calculate a signature of the current waveform; and determining whether the signature is greater than a predetermined signature threshold.
7. The method as set forth in claim 6, wherein if it is determined that the signature is greater than the predetermined signature threshold, recording an indication of a fault in memory.
8. The method as set forth in claim 6, wherein if it is determined that the signature is greater than the predetermined signature threshold, providing a notification that a fault has been detected.
9. The method as set forth in claim 6, further comprising the steps of: monitoring a number of times that the signature is greater than the predetermined signature threshold; and if the number exceeds a count threshold, providing a notification that a fault has been detected.
10. The method as set forth in claim 6, further comprising the step of filtering noise from the current waveform prior to calculating the signature of the current waveform.
11. A method for detecting a fault in a gear system for repositioning a seat assembly in an automotive vehicle, wherein the gear system includes a gear train operatively coupled to a motor, the method comprising the steps of: applying a voltage pulse to the motor; measuring a current drawn by the motor while the voltage pulse is applied to the motor; recording the current over time as a current waveform; determining a rise time in the current waveform for the current to reach a predetermined current amplitude; and determining whether the rise time is greater than a predetermined rise time threshold.
12. The method as set forth in claim 11, wherein if it is determined that the rise time is greater than the predetermined rise time threshold, recording an indication of a fault in memory.
13. The method as set forth in claim 11, wherein if it is determined that the rise time is greater than the predetermined rise time threshold, providing a notification that a fault has been detected.
14. The method as set forth in claim 11, further comprising the steps of: monitoring a number of times that the rise time is greater than the predetermined rise time threshold; and if the number exceeds a count threshold, providing a notification that a fault has been detected.

15. The method as set forth in claim 11, further comprising the step of filtering noise from the current waveform prior to calculating the rise time in the current waveform.
 16. A method for detecting a fault in a gear system for repositioning a seat assembly in an automotive vehicle, wherein the gear system includes a gear train operatively coupled to a motor having a drive shaft, the method comprising the steps of: measuring a hall effect signal from the drive shaft indicating a rotational position and a rotational speed of the drive shaft over time; comparing the hall effect signal to an expected hall effect signal for the motor; and determining whether the hall effect signal is different from the expected hall effect signal.
 17. The method as set forth in claim 16, wherein if it is determined that the hall effect signal is different from the expected hall effect signal, recording an indication of a fault in memory.
 18. The method as set forth in claim 16, wherein if it is determined that the hall effect signal is different from the expected hall effect signal, providing a notification that a fault has been detected.
 19. The method as set forth in claim 16, further comprising the steps of: monitoring a number of times that the hall effect signal is different from the expected hall effect signal; and if the number exceeds a count threshold, providing a notification that a fault has been detected.
 20. The method as set forth in claim 16, wherein the comparing step comprises determining whether a pulse period in the hall effect signal fluctuates over time.
 21. The method as set forth in claim 16, further comprising the step of terminating power to the motor, wherein the comparing step includes determining a number of pulses within the hall effect signal after terminating power to the motor.
 22. The method as set forth in claim 16, further comprising the step of determining whether the gear train has encountered an impact, wherein the comparing step includes determining a number of pulses within the hall effect signal after the gear train encountered the impact.
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